

A finite-rank obstruction to global triangular factorization in $GL^p(\mathcal{H})$

Candidate counterexample packet for arXiv:2511.02107

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Abstract

For every $1 < p < \infty$, the multiplication map underlying the upper/lower triangular factorization in Goliński–Tumpach Remark 7.4 is not surjective on $GL^p(\mathcal{H})$. A rotation through $\pi/2$ on the first two basis vectors, extended by the identity, is an explicit counterexample. It lies in every Schatten restricted group and even in its identity component. Nearby rotations admit explicit triangular factorizations, whose factors blow up at the exceptional angle; this pinpoints why the local factorization in Proposition 7.3 cannot be globalized.

Status and scope. This is a candidate full negative answer to the global triangular factorization question in source Remark 7.4. It does not address the different Iwasawa-decomposition question for restricted groups on polarized Hilbert spaces in source Remark 6.6.

1 Setting

Let $\mathcal{H}$ be a complex separable Hilbert space with orthonormal basis $(e_n)_{n \geq 1}$, and let $\mathcal{S}^p$ be the Schatten ideal. The source writes

$$GL^p(\mathcal{H}) = (\mathbf{I} + \mathcal{S}^p) \cap GL(\mathcal{H}).$$

Its strictly upper and lower-plus-diagonal Lie algebras are

$$\begin{aligned}\mathcal{S}_{++}^p &= \{X \in \mathcal{S}^p : X e_n \in \text{span}(e_1, \dots, e_{n-1})\}, \\ \mathcal{S}_-^p &= \{Y \in \mathcal{S}^p : Y e_n \in \overline{\text{span}}(e_n, e_{n+1}, \dots)\}.\end{aligned}$$

Thus the factorization used in source Proposition 7.3 has the form

$$A = g_+^{-1} g_-, \quad g_+ \in G_+ := (\mathbf{I} + \mathcal{S}_{++}^p) \cap GL(\mathcal{H}), \quad g_- \in G_- := (\mathbf{I} + \mathcal{S}_-^p) \cap GL(\mathcal{H}). \quad (1)$$

It exists locally near the identity. Remark 7.4 asks whether it exists globally on $GL^p(\mathcal{H})$.

2 Proof intuition

The first vector is rigid under every unit upper-triangular factor: $g_+ e_1 = e_1$. The second vector is constrained under every lower triangular factor: $g_- e_2$ cannot have an e_1 -component. An operator that sends e_2 to $-e_1$ makes these two requirements collide immediately. Extending this two-dimensional operator by the identity keeps it inside every Schatten restricted group, so infinite tail coordinates cannot supply a hidden pivot.

3 Counterexample and sharp path

Theorem 1. *For every $1 < p < \infty$, the map*

$$G_+ \times G_- \longrightarrow GL^p(\mathcal{H}), \quad (g_+, g_-) \longmapsto g_+^{-1}g_-$$

is not surjective. This remains false if the upper factor is allowed an arbitrary invertible diagonal.

Proof. Let A act on $E = \text{span}(e_1, e_2)$ by

$$A|_E = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

in the ordered basis (e_1, e_2) , and let A be the identity on $E^\perp$. It is unitary and $A - I$ has rank two. Therefore $A \in GL^p(\mathcal{H})$ for every finite p .

Suppose (1) held. Then $g_- = g_+A$. Strict upper triangularity gives $g_+e_1 = e_1$, whereas $Ae_2 = -e_1$. Consequently

$$g_-e_2 = g_+Ae_2 = -g_+e_1 = -e_1. \quad (2)$$

But by lower triangularity,

$$g_-e_2 \in \overline{\text{span}}(e_2, e_3, \dots),$$

contradicting (2).

If the upper triangular group is enlarged to allow arbitrary invertible diagonal entries, then $g_+e_1 = ce_1$ for some $c \neq 0$, and the same argument gives $g_-e_2 = -ce_1$, again impossible. $\square$

The counterexample is not separated from the identity by a component obstruction. Define the smooth finite-rank rotation path

$$A_\theta|_E = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad A_\theta|_{E^\perp} = I. \quad (3)$$

Theorem 2. *If $\cos \theta \neq 0$, then A_θ has a factorization of the form (1). One may take, on E ,*

$$g_+(\theta) = \begin{pmatrix} 1 & \tan \theta \\ 0 & 1 \end{pmatrix}, \quad g_-(\theta) = \begin{pmatrix} \sec \theta & 0 \\ \sin \theta & \cos \theta \end{pmatrix}, \quad (4)$$

with both operators equal to the identity on $E^\perp$. At $\theta = \pi/2$ no such factorization exists, and the displayed factors blow up as $\theta \rightarrow \pi/2$.

Proof. Both factors in (4) are invertible finite-rank perturbations of identity and have the required triangular forms. Direct multiplication using $\sin^2 \theta + \cos^2 \theta = 1$ gives

$$g_+(\theta)A_\theta = g_-(\theta),$$

hence $A_\theta = g_+(\theta)^{-1}g_-(\theta)$. Moreover $\|g_+(\theta) - I\|_p = |\tan \theta|$, while the upper-left diagonal entry of $g_-(\theta)$ is $\sec \theta$. Both are unbounded at $\pi/2$, and Theorem 1 proves nonexistence at the endpoint. $\square$

Corollary 1. *The local factorization chart containing the identity cannot be continued globally along the path A_θ , even though that path consists entirely of finite-rank perturbations of identity.*

Convention check. The same endpoint also fails a factorization in the reverse order $A = LU$ with invertible lower and upper triangular factors. Indeed, $Ue_1 = ce_1$ with $c \neq 0$ would force $Le_1 = c^{-1}e_2$. Lower triangularity then makes every entry in the first row of L zero, so L is not surjective. The negative answer is therefore independent of whether the phrase “LU factorization” is attached to the source’s displayed upper–lower order or to the reverse naming convention.

4 Source evidence

The source page below states the local factorization in Proposition 7.3 and the global question in Remark 7.4.

Proposition 7.3. *For $1 < p < +\infty$, consider the decomposition $L^p(\mathcal{H}) = L^p(\mathcal{H})_- \oplus L^p(\mathcal{H})_{++}$ with $\text{Ad}_{GL^p(\mathcal{H})}$ -invariant non-degenerate symmetric bilinear continuous map given by the trace*

$$\langle A, B \rangle = \text{Tr}(AB), A, B \in L^p(\mathcal{H}). \quad (7.6)$$

Let $R = p_{L^p(\mathcal{H})_{++}} - p_{L^p(\mathcal{H})_-}$ be the associated R -matrix with $p_{L^p(\mathcal{H})_{++}}$ and $p_{L^p(\mathcal{H})_-}$ the projections on $L^p(\mathcal{H})_{++}$ and $L^p(\mathcal{H})_-$ with respect to the previous decomposition of $L^p(\mathcal{H})$. Consider the family of spectral functions

$$F_k(\mu) = \frac{1}{k+1} \text{Tr} \mu^{k+1}, k \in \mathbb{N}, \mu \in L^p(\mathcal{H}). \quad (7.7)$$

Then we have:

- (1) $\{F_i, F_j\}_R = 0, \forall i, j \in \mathbb{N}$
- (2) *the flow of the Hamiltonian vector field $X_{F_k} := \{\cdot, F_k\}_R$ associated with F_k with respect to the Poisson bracket $\{\cdot, \cdot\}_R$ satisfies the Lax equation*

$$\frac{d\mu}{dt} = X_{F_k}(\mu) = [\mu, p_{L^p(\mathcal{H})_{++}}(\mu^k)] = -[\mu, p_{L^p(\mathcal{H})_-}(\mu^k)]. \quad (7.8)$$

- (3) *the integral curve of the Hamiltonian vector field X_{F_k} , starting at $\mu_0 \in L^p(\mathcal{H})$, is given by*

$$\mu(t) = \text{Ad}_{g_+(t)} \mu_0 = \text{Ad}_{g_-(t)} \mu_0,$$

where $g_+(t) \in \mathbb{1} + L^p(\mathcal{H})_{++}$ and $g_-(t) \in \mathbb{1} + L^p(\mathcal{H})_-$ are the smooth curves solving the factorization problem for $|t|$ small enough

$$\exp(-t\mu_0^k) = g_+(t)^{-1}g_-(t), \text{ with initial conditions } g_{\pm}(0) = e, \quad (7.9)$$

Remark 7.4. In this case, we do not know if we have a global decomposition of $GL^p(\mathcal{H})$ into the product of the groups of upper and lower triangular operators. In finite-dimension, this is known as the LU -factorization. However, as mentioned above, the solution of the factorization problem exists at least locally.

Figure 1: Goliński–Tumpach, arXiv:2511.02107, PDF page 32 (cropped).

Reference

T. Goliński and A. B. Tumpach, *Banach Poisson–Lie groups, Lax equations and the AKS theorem in infinite dimensions*, arXiv:2511.02107, especially Proposition 7.3 and Remark 7.4.